

Properties of Regular Languages

For regular languages L_1 and L_2
we will prove that:

Union: $L_1 \cup L_2$

Concatenation: $L_1 L_2$

Star: L_1^*

Reversal: L_1^R

Complement: $\overline{L_1}$

Intersection: $L_1 \cap L_2$

Are regular
Languages

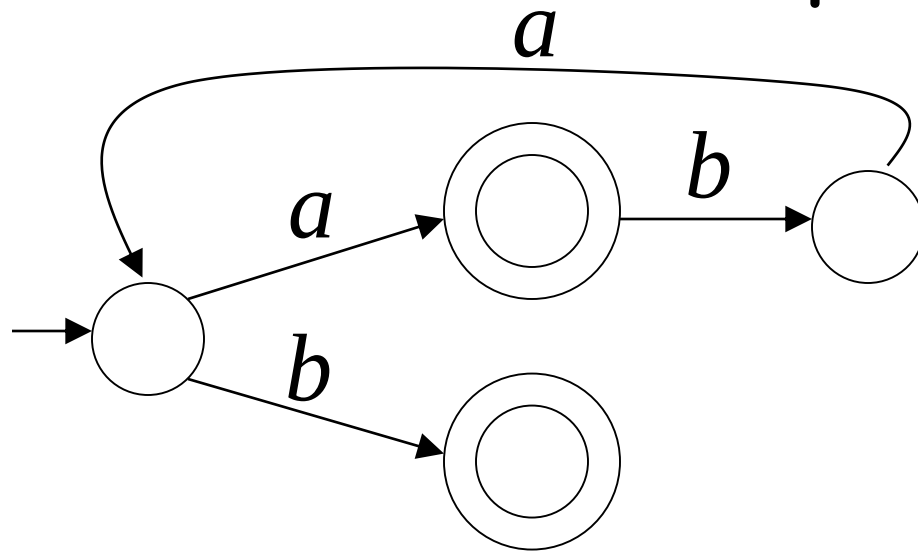
A useful transformation: use one accept state

Any NFA can be converted

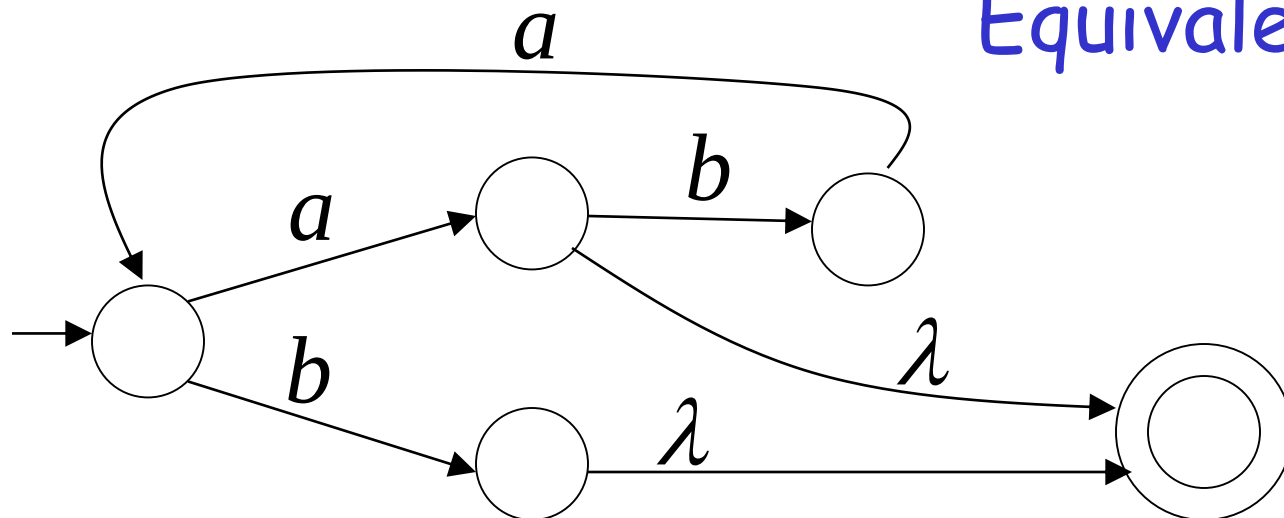
to an equivalent NFA

with a single final state

Example



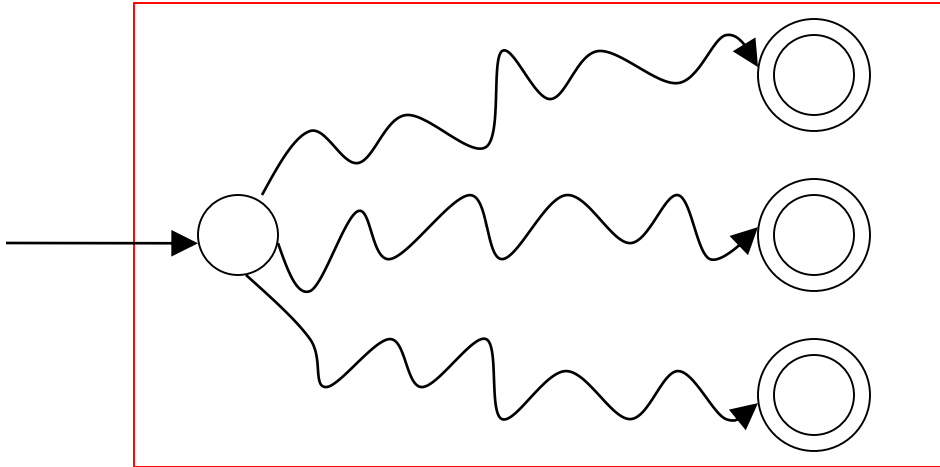
NFA



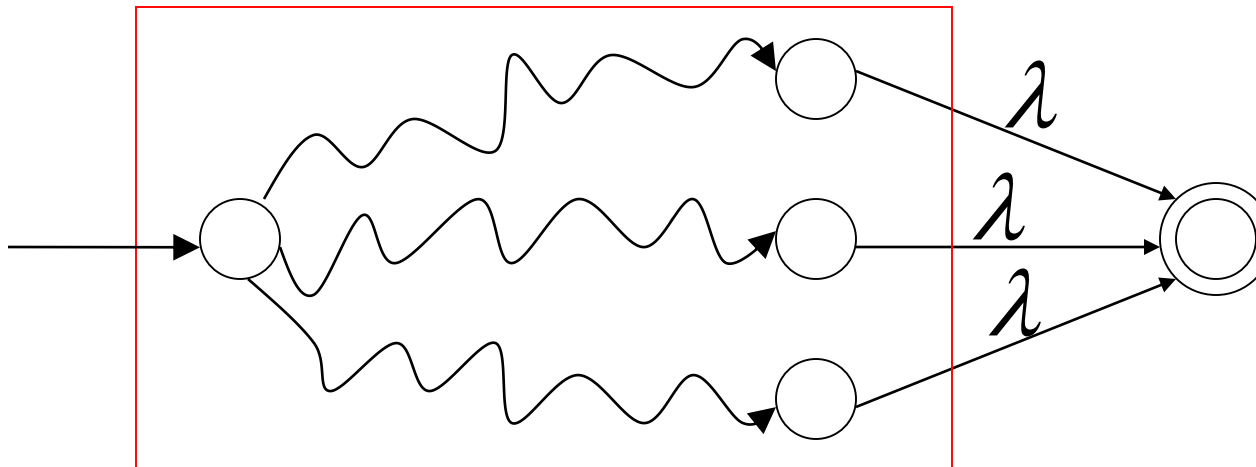
Equivalent NFA

In General

NFA



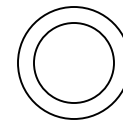
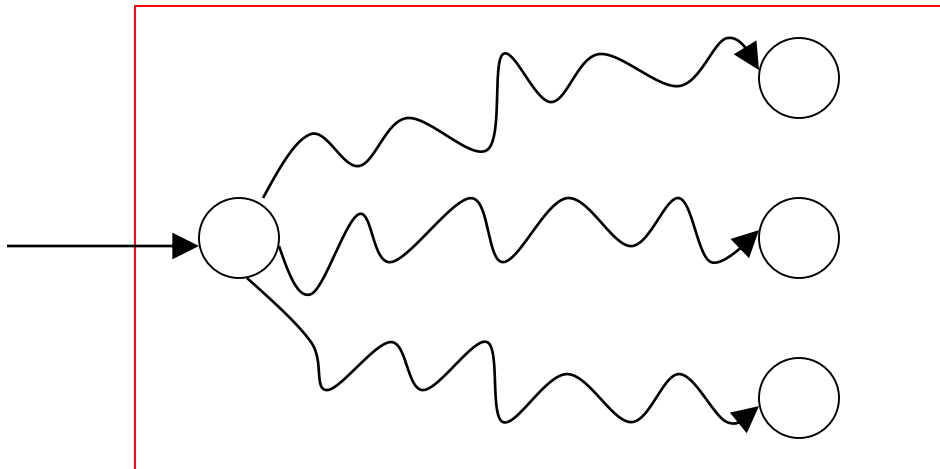
Equivalent NFA



Single
final state

Extreme Case

NFA without final state



Add a final state
Without transitions

We say: Regular languages are **closed under**

Union: $L_1 \cup L_2$

Concatenation: $L_1 L_2$

Star: L_1^*

Reversal: L_1^R

Complement: $\overline{L_1}$

Intersection: $L_1 \cap L_2$

Let us take two languages

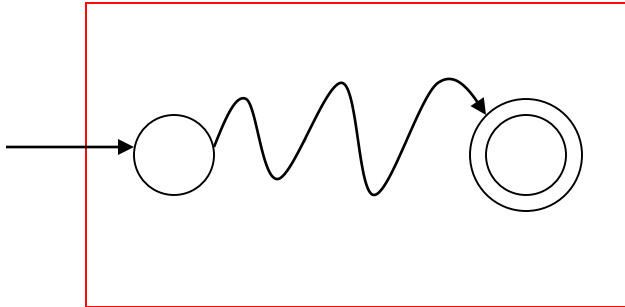
Regular language L_1

Regular language L_2

$$L(M_1) = L_1$$

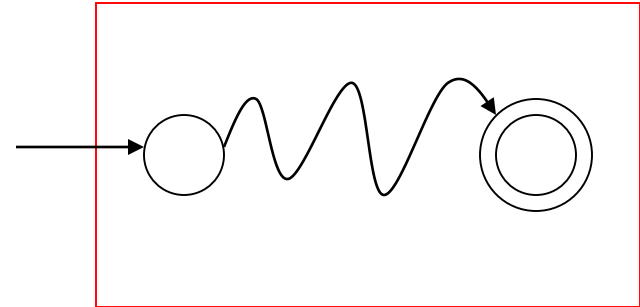
$$L(M_2) = L_2$$

NFA M_1



Single final state

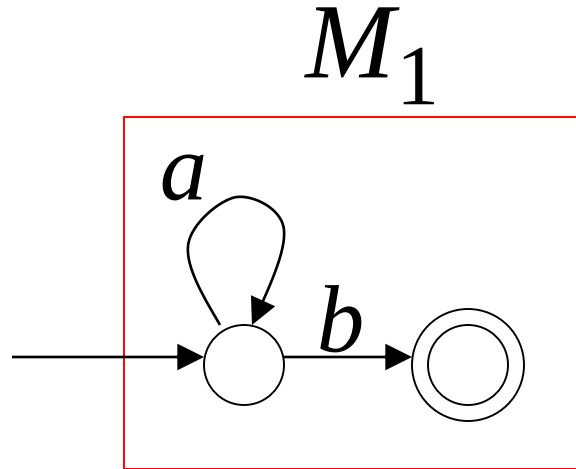
NFA M_2



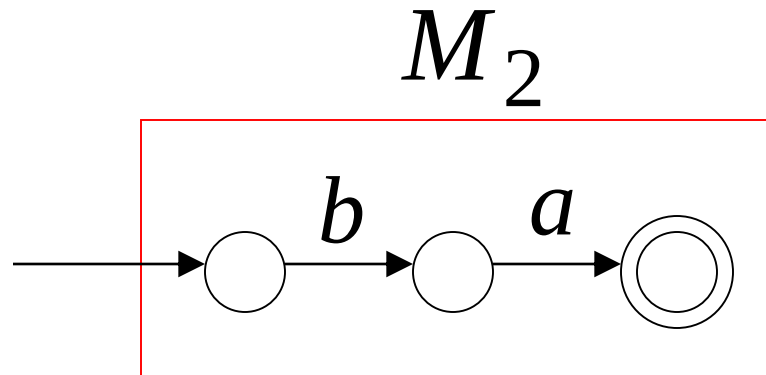
Single final state

Example

$$L_1 = \{a^n b \mid n \geq 0\}$$



$$L_2 = \{ba\}$$



Show Regular languages are **closed under**

Union: $L_1 \cup L_2$

Concatenation: $L_1 L_2$

Star: L_1^*

Reversal: L_1^R

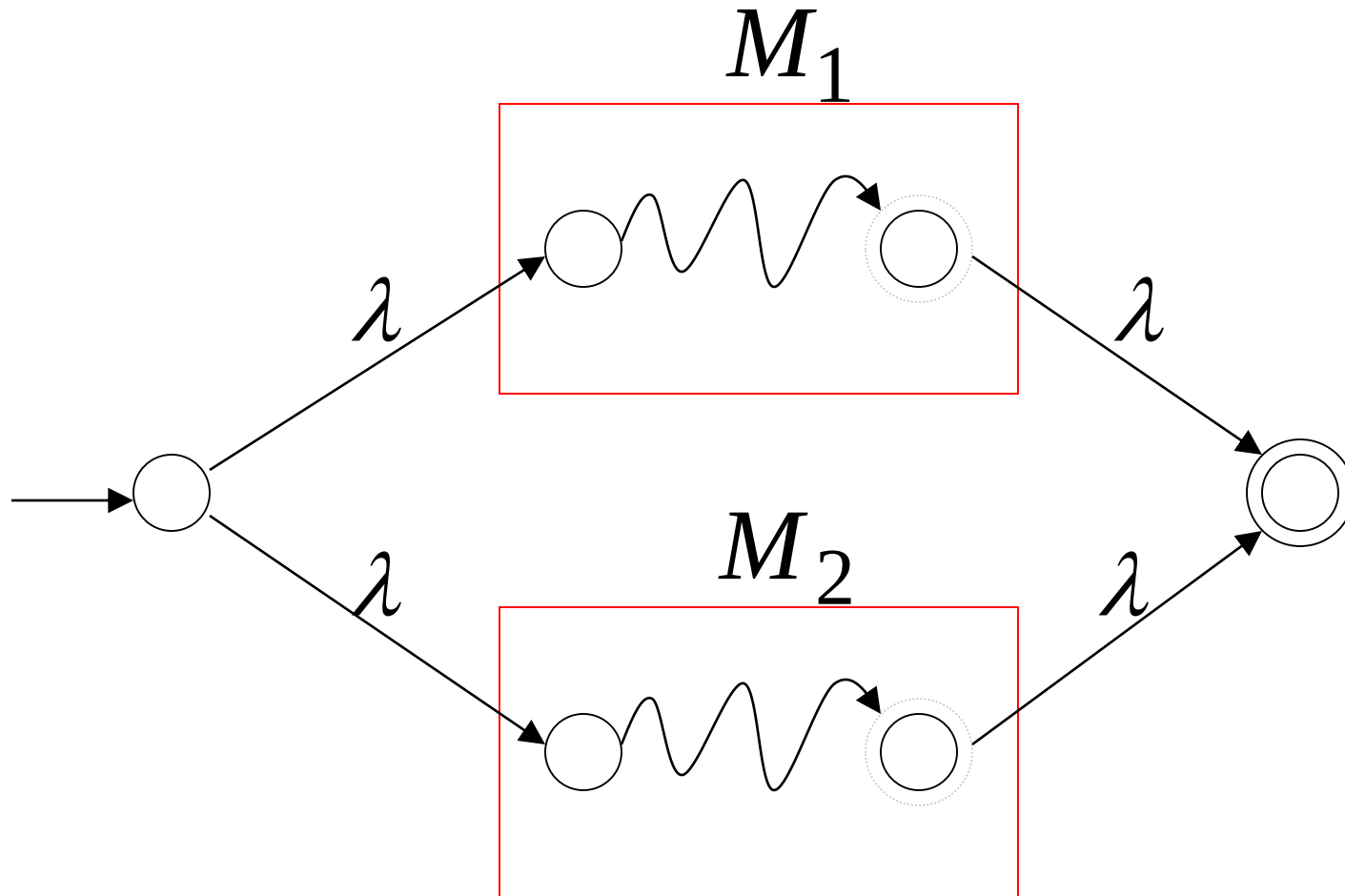
Complement: $\overline{L_1}$

Intersection: $L_1 \cap L_2$

HOW?

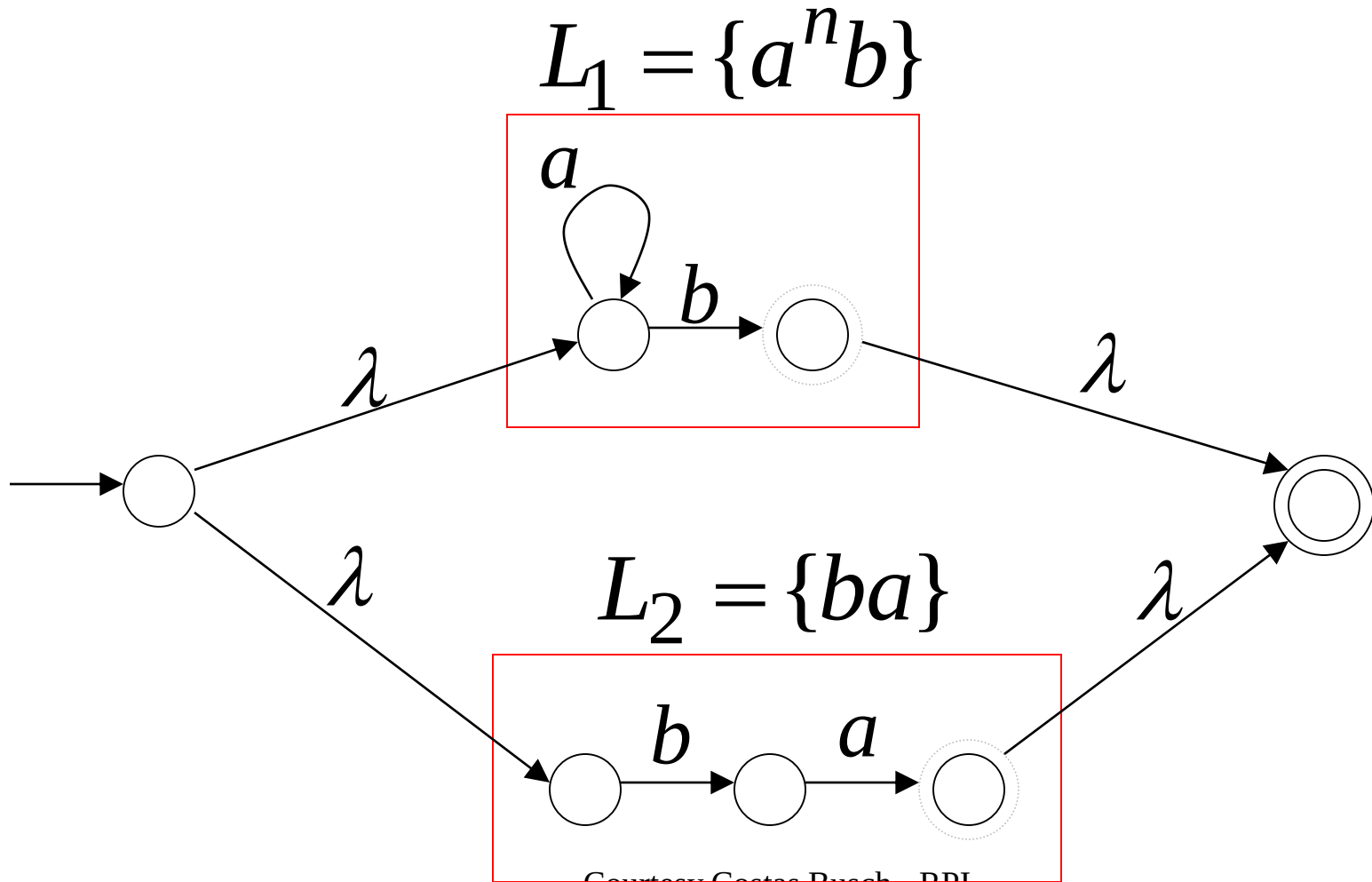
Union

NFA for $L_1 \cup L_2$



Example

NFA for $L_1 \cup L_2 = \{a^n b\} \cup \{ba\}$



Show Regular languages are **closed under**

Union: $L_1 \cup L_2$

Concatenation: $L_1 L_2$

Star: L_1^*

Reversal: L_1^R

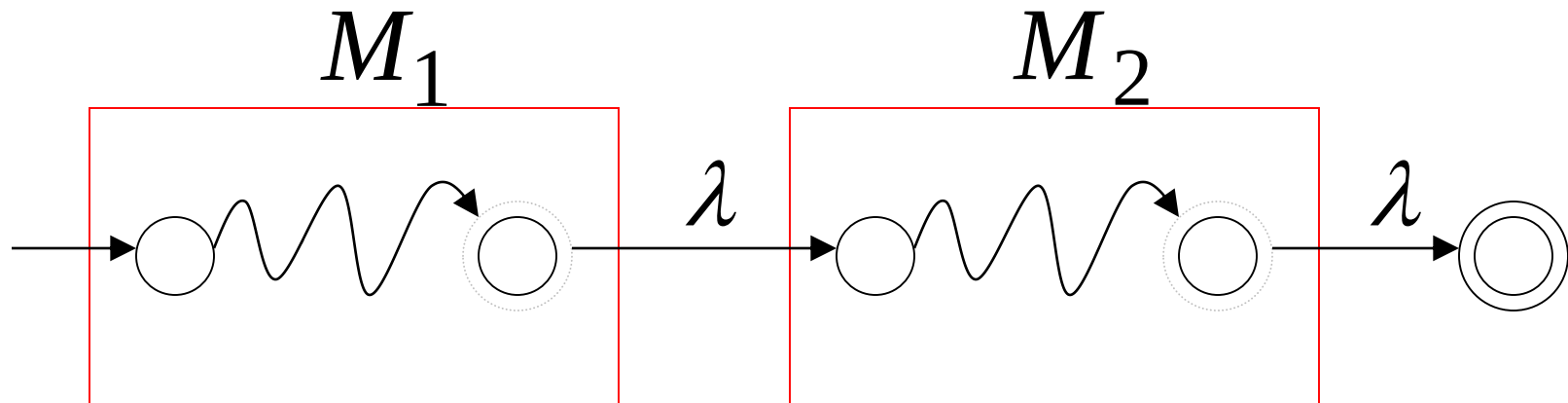
Complement: $\overline{L_1}$

Intersection: $L_1 \cap L_2$

HOW?

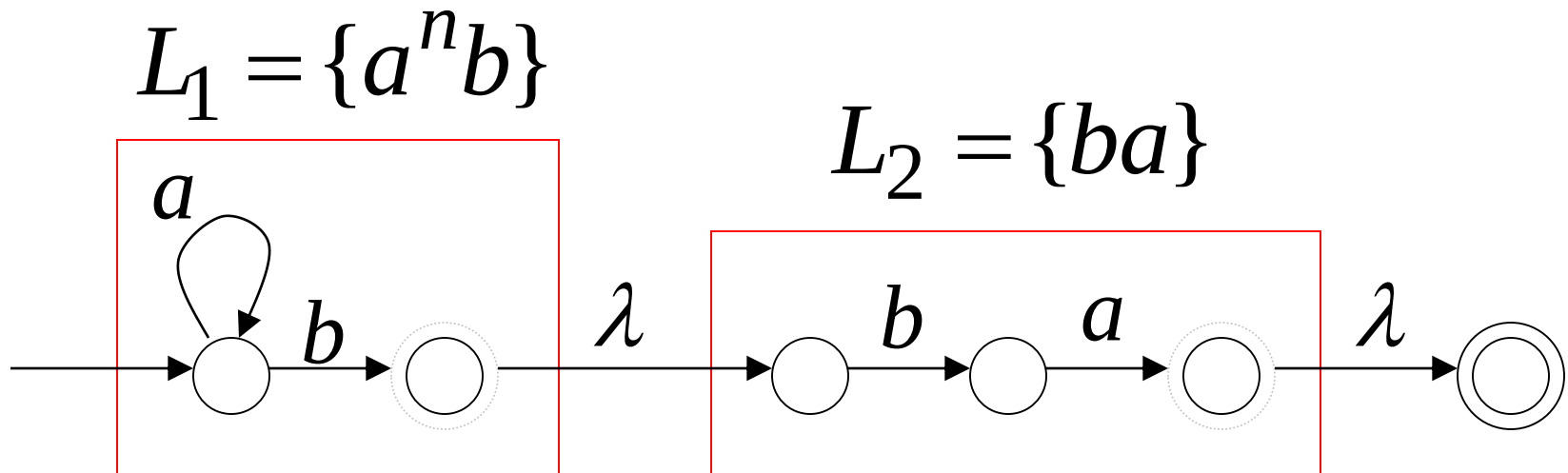
Concatenation

NFA for L_1L_2



Example

NFA for $L_1L_2 = \{a^n b\} \{ba\} = \{a^n bba\}$



Show Regular languages are **closed under**

Union: $L_1 \cup L_2$

Concatenation: $L_1 L_2$

Star: L_1^*

HOW?

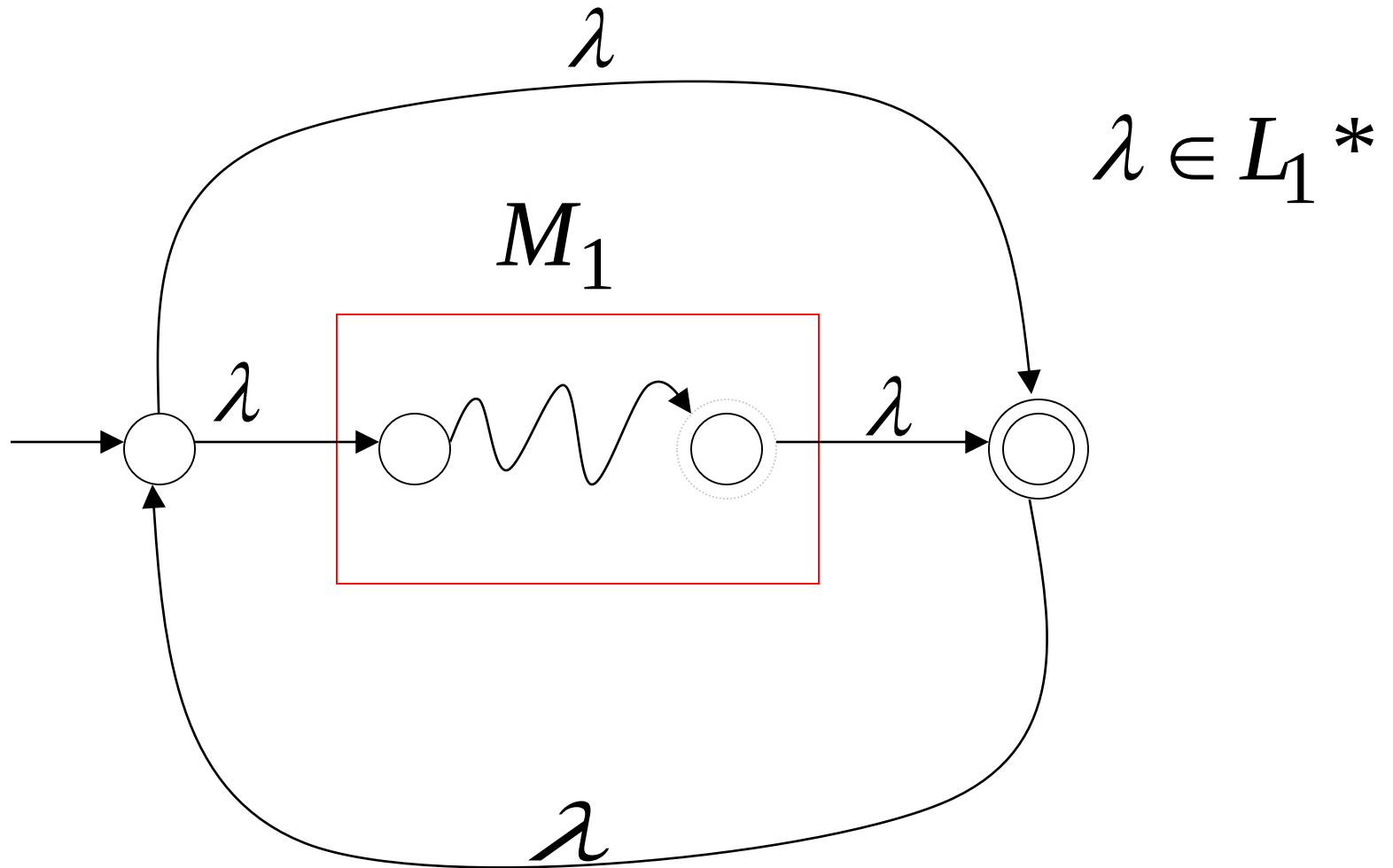
Reversal: L_1^R

Complement: $\overline{L_1}$

Intersection: $L_1 \cap L_2$

Star Operation

NFA for L_1^*

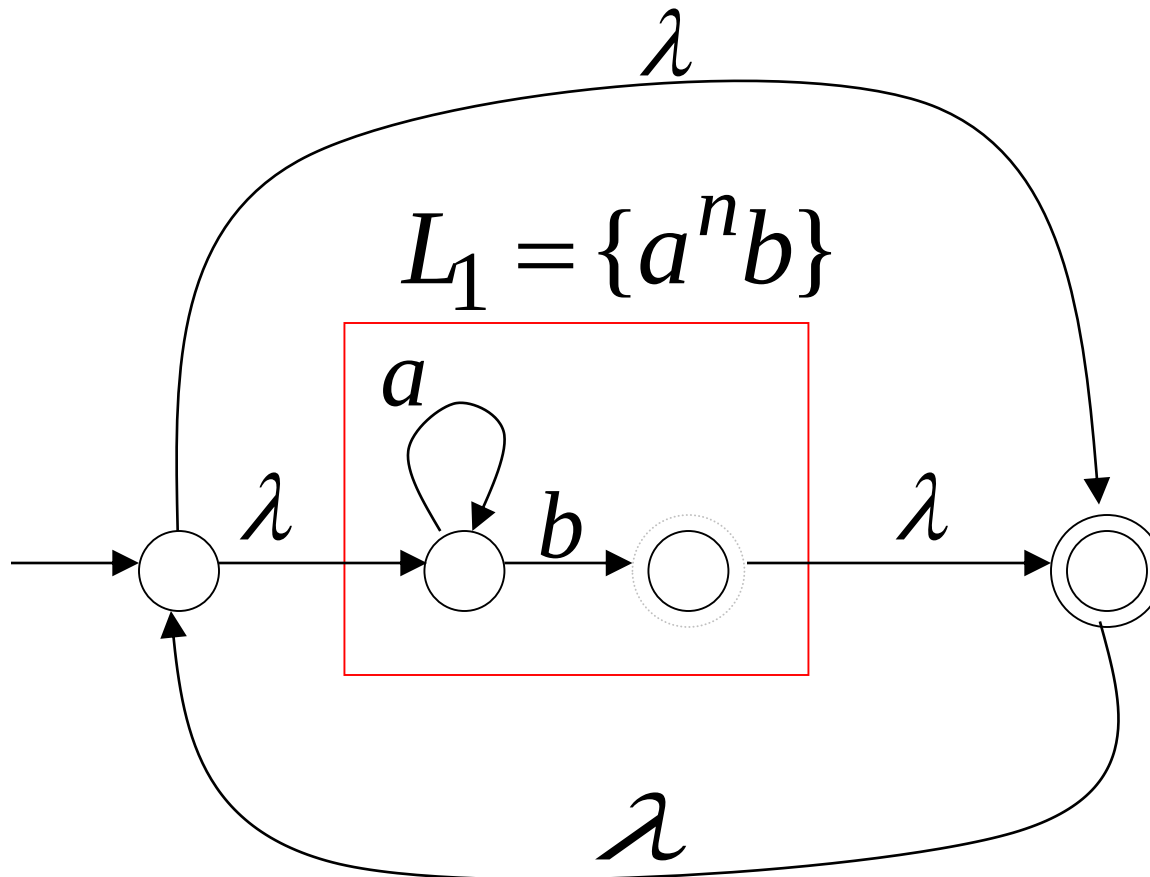


Example

NFA for $L_1^* = \{a^n b\}^*$

$$w = w_1 w_2 \cdots w_k$$

$$w_i \in L_1$$



Show Regular languages are **closed under**

Union: $L_1 \cup L_2$

Concatenation: $L_1 L_2$

Star: L_1^*

Reversal: L_1^R

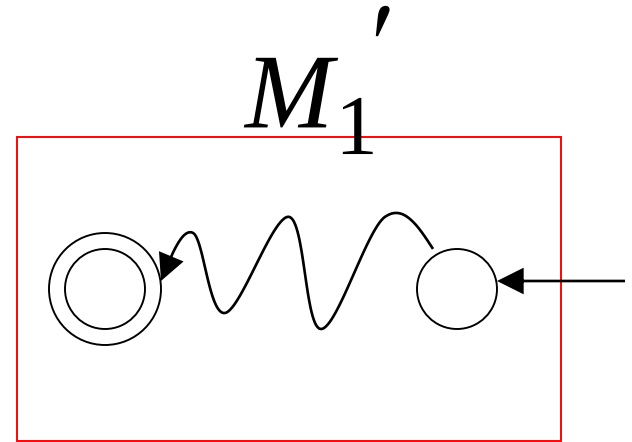
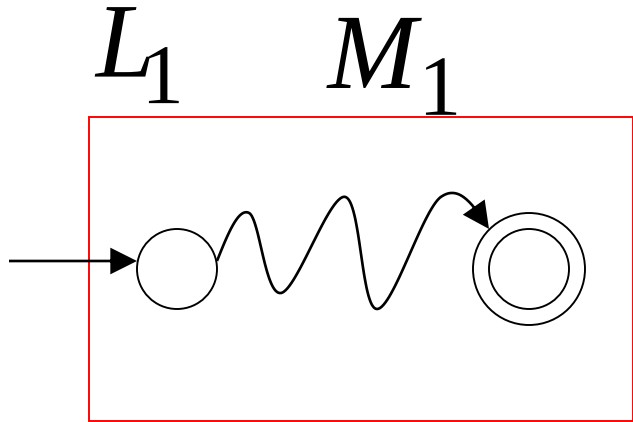
HOW?

Complement: $\overline{L_1}$

Intersection: $L_1 \cap L_2$

Reverse

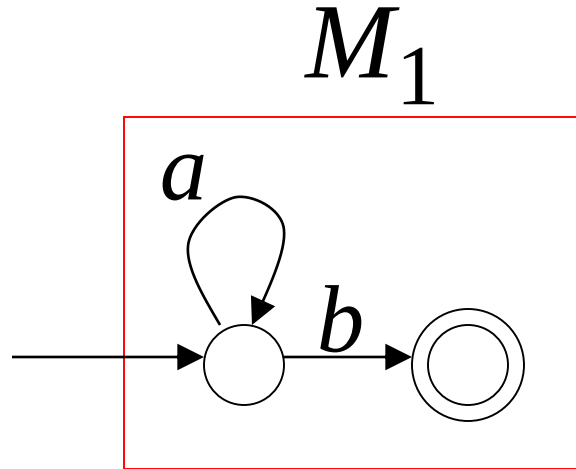
NFA for L_1^R



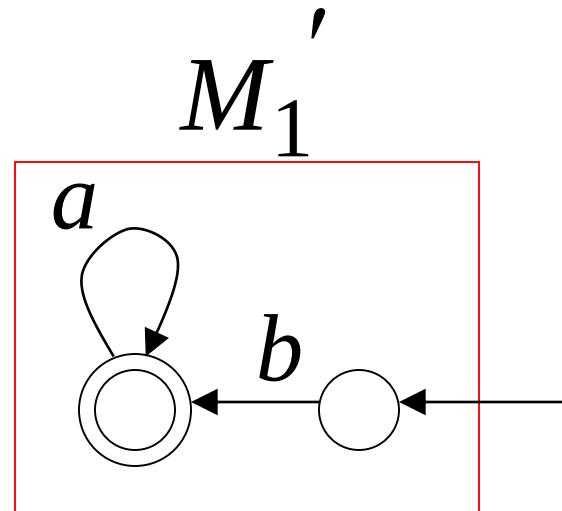
1. Reverse all transitions
2. Make initial state final state and vice versa

Example

$$L_1 = \{a^n b\}$$



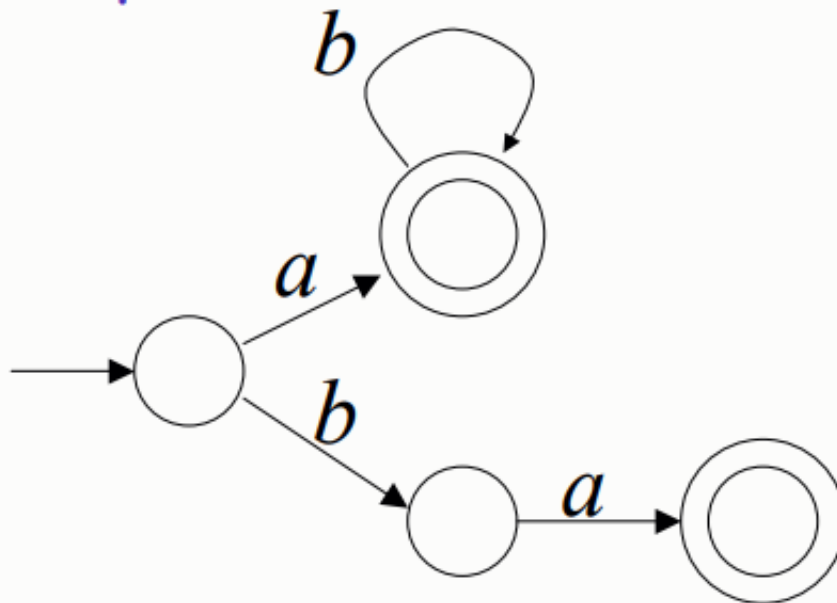
$$L_1^R = \{ba^n\}$$



Proof

Since L is regular,
there is NFA that accepts L

Example:



$$L = ab^* + ba$$

Show Regular languages are **closed under**

Union: $L_1 \cup L_2$

Concatenation: $L_1 L_2$

Star: L_1^*

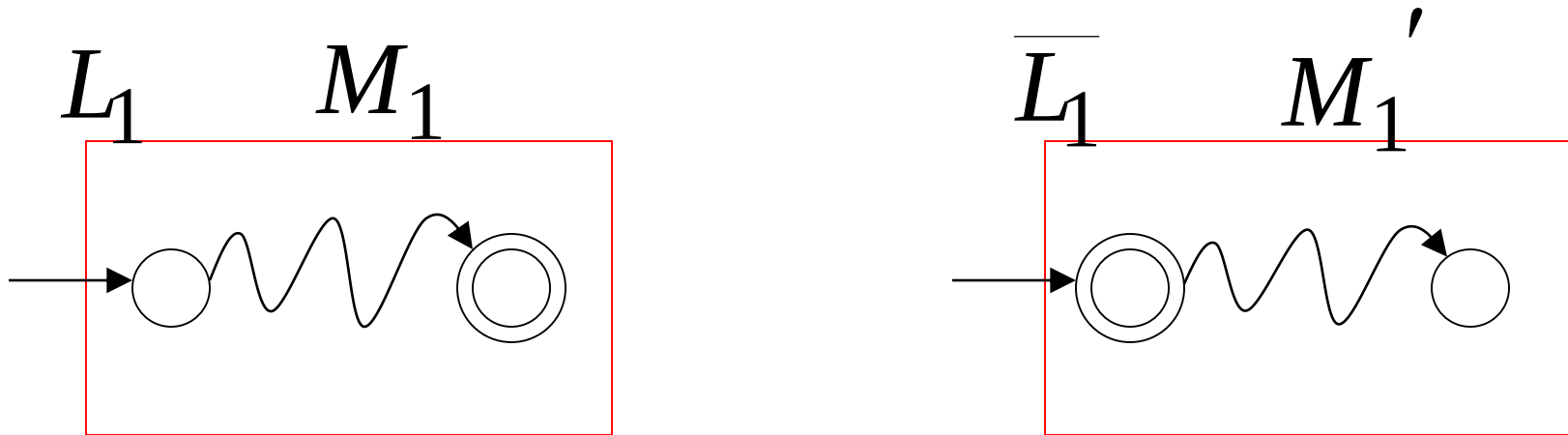
Reversal: L_1^R

HOW?

Complement: $\overline{L_1}$

Intersection: $L_1 \cap L_2$

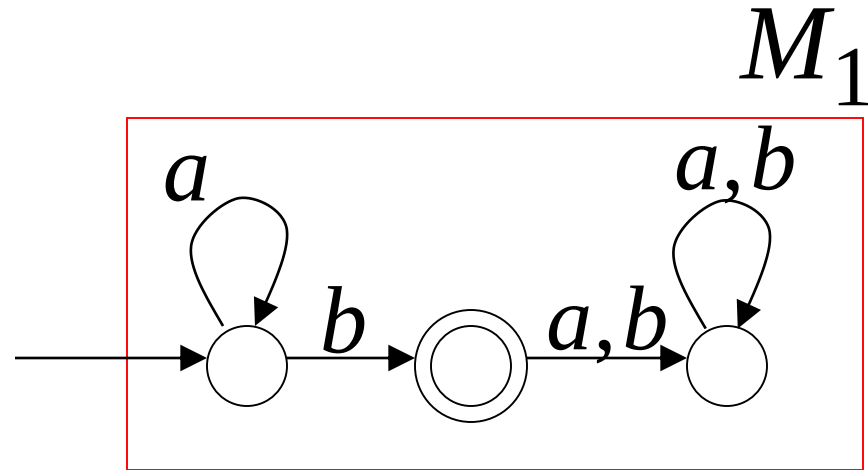
Complement



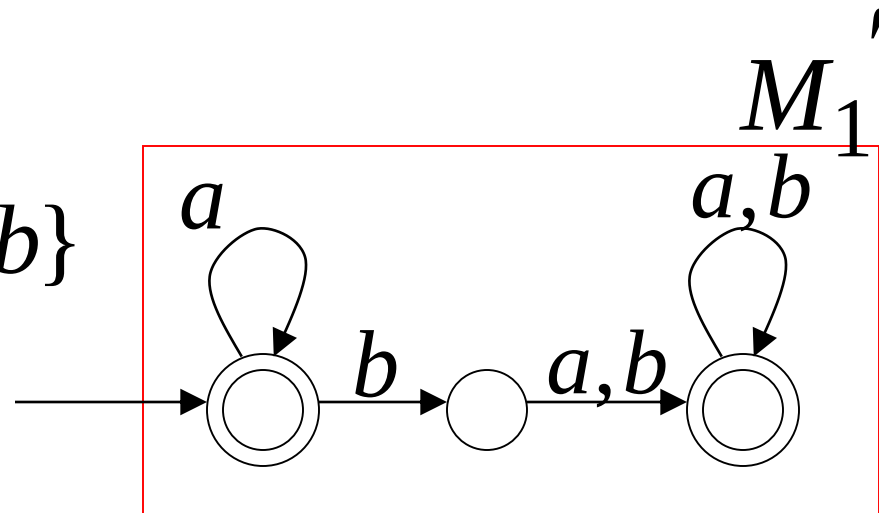
1. Take the **DFA** that accepts L_1
2. Make final states non-final, and vice-versa

Example

$$L_1 = \{a^n b\}$$



$$\overline{L_1} = \{a,b\}^* - \{a^n b\}$$



Show Regular languages are **closed under**

Union: $L_1 \cup L_2$

Concatenation: $L_1 L_2$

Star: L_1^*

Reversal: L_1^R

Complement: $\overline{L_1}$

Intersection: $L_1 \cap L_2$

HOW?

Intersection

L_1 regular



L_2 regular

We show

$L_1 \cap L_2$
regular

Proof for Intersection Closure

Machine M_1

DFA for L_1

Machine M_2

DFA for L_2

Construct a new DFA M that accepts $L_1 \cap L_2$

M simulates in parallel M_1 and M_2

M simulates in parallel M_1 and M_2

M accepts string w if and only if:

M_1 accepts string w
and M_2 accepts string w

$$L(M) = L(M_1) \cap L(M_2)$$

Another Proof for Intersection Closure

DeMorgan's Law: $L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}}$

L_1, L_2 regular

→ $\overline{L_1}, \overline{L_2}$ regular

→ $\overline{L_1} \cup \overline{L_2}$ regular

→ $\overline{\overline{L_1} \cup \overline{L_2}}$ regular

→ $L_1 \cap L_2$ regular

Example

$$\left. \begin{array}{l} L_1 = \{a^n b\} \text{ regular} \\ L_2 = \{ab, ba\} \text{ regular} \end{array} \right\} \Rightarrow L_1 \cap L_2 = \{ab\} \text{ regular}$$